

A reaction was performed in which compounds A and B were combined to form compound C. The table below shows an analysis of the kinetics for the reaction with changing reagent concentration.

Trial	[A] (M)	[B] (M)	Initial rate (M/s)
1	0.18	0.18	4.65×10^{-3}
2	0.18	0.36	9.30×10^{-3}
3	0.36	0.18	9.30×10^{-3}
4	0.36	0.36	1.86×10^{-2}

Based on the kinetic data shown in the table, which overall reaction order and rate law best describe this reaction?

- ☐ A. First-order: Rate = $k[A]$ [2%]
☐ B. First-order: Rate = $k[B]$ [8%]
☒ C. Second-order: Rate = $k[A][B]$ [84%]
☐ D. Fourth-order: Rate = $k[A][B]$ [3%]

Correct

83% answered correctly

Explanation:

Reaction equation		Reaction rate law	
A + B → C		Rate = $k[A]^m[B]^n$ → Rate = $k[A][B]$ $m + n$ = overall reaction order → $1 + 1 = 2$ (2 nd order)	
Trial	[A] (M)	[B] (M)	Initial rate (M/s)
1	0.18	0.18	4.65×10^{-3}
2	0.18	0.36	9.30×10^{-3}
3	0.36	0.18	9.30×10^{-3}
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Rate has first-order dependence upon compound B ($n = 1$)
 Rate has first-order dependence upon compound A ($m = 1$)

The **rate law** of a reaction relates the rate of reaction to the concentration of each reactant. When two reactants are present, the rate law is expressed generally by the equation:

$$\text{Rate} = k[A]^m[B]^n$$

where k is a constant, m and n are the order of the respective reactants, and $(m + n)$ is the **overall reaction order**. To **determine** the actual **rate law**, the values of m and n must be found by **comparing** the given **reactant concentrations** and **initial rates**.

Comparing Trial 2 to Trial 1, [A] is left unchanged but [B] is doubled relative to Trial 1, causing the reaction rate to double. Therefore, the rate has a first-order dependence upon [B], and $n = 1$. Similarly, when comparing Trial 3 to Trial 1, [A] is doubled but [B] is left unchanged, and the rate doubles; so, the reaction rate also has a first-order dependence upon [A], and $m = 1$.

This result is further verified by comparing Trial 4 with Trial 1; doubling both [A] and [B] gives a rate four times greater than that seen for Trial 1.

Therefore, the overall reaction rate must be a second-order expression ($m + n = 2$) involving variable terms for both [A] and [B].

(Choices A and B) These are rate laws that depend on only one chemical species but not the other (either m or n equals zero). However, the data show that the rate depends on both [A] and [B].

(Choice D) A fourth-order reaction ($m + n = 4$) is inconsistent with the second-order ($m + n = 2$) rate law expression given. In a fourth-order reaction (i.e., $\text{Rate} = k[\text{A}][\text{B}]^3$, $\text{Rate} = k[\text{A}]^3[\text{B}]$, or $\text{Rate} = k[\text{A}]^2[\text{B}]^2$), kinetic data would show a four-fold or an eight-fold rate increase upon doubling the reagent with the higher order.

Educational objective:

Kinetic data can be used to determine rate laws and reaction orders. If doubling a reagent's concentration doubles the reaction rate, the reaction is first-order with respect to that reagent. If doubling the concentration quadruples the rate, the reaction is second-order with respect to the reagent, and so on. The overall order of a reaction is the sum of the reaction orders of each rate-influencing reagent.

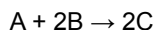
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Subject:
General Chemistry

QID: 401423

System:
Thermodynamics, Kinetics & Gas Laws

A combination reaction was performed as described by the reaction:



Analysis of the kinetics for the reaction found that the reaction progress responded to changes in reagent concentration as given in the following table:

Trial	[A] (M)	[B] (M)	Initial rate (M/s)
1	2.00	0.25	0.50
2	2.00	0.50	2.00
3	3.00	0.50	3.00
4	1.00	0.50	1.00

Based on the kinetic data, which of the following expressions gives the rate constant for the rate law of this reaction?

- ☐ A. $k = \frac{\text{Rate}}{[\text{A}][\text{B}]^3}$ [3%]
- ☒ B. $k = \frac{\text{Rate}}{[\text{A}][\text{B}]^2}$ [88%]
- ☐ C. $k = \frac{\text{Rate}}{[\text{A}]^2[\text{B}]}$ [4%]
- ☐ D. $k = \frac{\text{Rate}}{[\text{A}][\text{B}]}$ [3%]

Correct

87% answered correctly

Explanation:

Reaction equation		Reaction rate law	
$\text{A} + 2\text{B} \rightarrow 2\text{C}$		$\text{Rate} = k[\text{A}]^m[\text{B}]^n \rightarrow k = \frac{\text{Rate}}{[\text{A}]^m[\text{B}]^n}$	
Trial	[A] (M)	[B] (M)	Initial rate (M/s)
1	2.00	0.25	0.50
2	2.00	0.50 $2 \times$	2.00 $4 \times$
3	3.00 $3 \times$	0.50	3.00 $3 \times$
4	1.00	0.50	1.00

Rate has second-order dependence upon compound B ($n = 2$)

Rate has first-order dependence upon compound A ($m = 1$)

The general **rate law** for a reaction is given by the equation:

$$\text{Rate} = k[\text{A}]^m[\text{B}]^n$$

The values of m and n , which give the **reaction order**, can be **determined** by **comparing** the reactant concentrations and the initial rates of given **kinetic data**. The rate law can then be determined, and the **rate constant** k can be **calculated** by solving for k :

$$k = \frac{\text{Rate}}{[\text{A}]^m[\text{B}]^n}$$

Comparing Trial 2 to Trial 1, [A] is left unchanged but [B] is doubled and the reaction rate quadruples. This means that the rate has second-order dependence on [B] ($n = 2$). Comparing Trial 3 to Trial 4, [A] is tripled but [B] is left unchanged. The rate triples, indicating that the reaction rate has first-order dependence on [A] ($m = 1$). Therefore, the overall reaction is third-order ($m + n = 3$), and the rate law can be written as:

$$\text{Rate} = k[\text{A}][\text{B}]^2$$

(Choice A) This expression for k comes from the incorrect rate law, $\text{Rate} = k[\text{A}][\text{B}]^3$, which has third-order dependence on [B] ($n = 3$) rather than second-order dependence ($n = 2$). Third-order dependence would cause the reaction rate to be eight times greater when [B] is doubled.

(Choice C) This expression for k comes from the incorrect rate law, $\text{Rate} = k[\text{A}]^2[\text{B}]$, which has second-order dependence on [A] ($m = 2$) and first-order dependence on [B] ($n = 1$). This would cause the reaction rate to be four times greater when [A] is doubled and only two times greater when [B] is doubled.

(Choice D) This expression for k comes from the incorrect rate law, $\text{Rate} = k[\text{A}][\text{B}]$, which has first-order dependence on [B] ($n = 1$). This would cause the reaction rate only to double when [B] is doubled.

Educational objective:

The reaction order of each reactant in a chemical process can be determined by comparing kinetic data at different concentrations, from which the rate law can be written. The rate constant k can then be evaluated from the rate law.

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